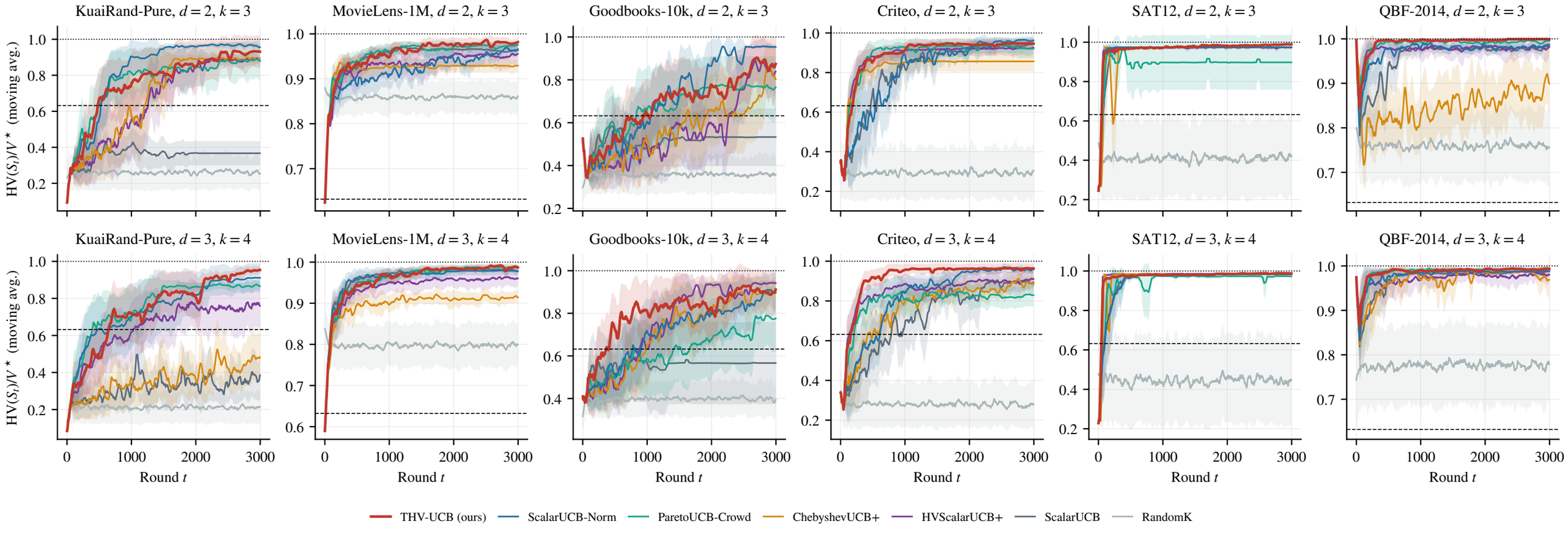

KuaiRand-Pure, $d = 2$, $k = 3$
MovieLens-1M, $d = 2$, $k = 3$
Goodbooks-10k, $d = 2$, $k = 3$
Criteo, $d = 2$, $k = 3$
SAT12, $d = 2$, $k = 3$
QBF-2014, $d = 2$, $k = 3$
KuaiRand-Pure, $d = 3$, $k = 4$
MovieLens-1M, $d = 3$, $k = 4$
Goodbooks-10k, $d = 3$, $k = 4$
Criteo, $d = 3$, $k = 4$
SAT12, $d = 3$, $k = 4$
QBF-2014, $d = 3$, $k = 4$
$\mathrm{HV}(S_t)/V^\star$ (moving avg.)
Round $t$
THV-UCB (ours)
ScalarUCB-Norm
ParetoUCB-Crowd
ChebyshevUCB+
HVScalarUCB+
ScalarUCB
RandomK